

ISABELLA RIVERA

Toronto, ON, Canada | +1 (437)-226-9302 | isabella.rivera@mail.utoronto.ca | [linkedin.com/in/isabellariv](https://www.linkedin.com/in/isabellariv) | github.com/isabellamrivera | isabellamrivera.github.io

Education

Honours Bachelor of Science, University of Toronto — St. George Campus

Expected June 2027

Specialist in Astronomy and Physics; Minor in Mathematics

Research Experience

Radio Search for Off-Axis GRB Jets — Undergraduate Thesis

September 2026 – Present

Dunlap Institute for Astronomy & Astrophysics · Supervised by Prof. Maria Drout and Dr. Genevieve Schroeder

- Search VLASS data for late-time radio emission from off-axis relativistic jets associated with Type Ic supernovae.
- Model radio light curves to constrain jet energetics and viewing angles, and assess which supernovae may have hosted off-axis GRBs.

FRB Multi-wavelength Counterpart Searches — SURP

May 2026 – Present

Dunlap Institute for Astronomy & Astrophysics · Supervised by Prof. Maria Drout, Prof. Juan Mena-Parra, and Mattias Lazda

- Searching for persistent radio sources (PRSs) associated with fast radio bursts (FRBs) within the CHIME/FRB Collaboration.
- Use archival radio and optical surveys to identify candidate counterparts and characterize the local environments of FRBs.

CHIME/FRB Far-Sidelobe Localization — AST431 (A+)

Sept 2025 – June 2026

David A. Dunlap Department of Astronomy & Astrophysics · Supervised by Prof. Juan Mena-Parra

- Developed and tested localization techniques for FRBs detected in CHIME's far sidelobes by characterizing instrumental systematics and evaluating geometric corrections.
- Applied improved localizations to support probabilistic host-galaxy association and follow-up.

Tracing Star Formation Histories in Vela — AST430 (A+)

Jan 2026 – May 2026

David A. Dunlap Department of Astronomy & Astrophysics · Supervised by Dr. Ronan Kerr and Prof. Joshua Speagle (SPYGLASS program)

- Investigated star-forming regions in the Vela Complex, studying their morphology and evolutionary histories.
- Used Gaia astrometry, stellar age fitting, and orbital traceback to identify possible shared formation histories among young stellar populations.

Radio Astronomy Research Assistant

May 2025 – Sept 2025

David A. Dunlap Department of Astronomy & Astrophysics · Supervised by Prof. Juan Mena-Parra

- Developed and tested clock synchronization methods for the CHIME Outriggers using noise modelling, interpolation, and timing-error analysis.
- Explored a new localization approach for mitigating timing uncertainties between telescope stations.

Publications

Gusinskaia, N. + co-authors including Rivera, I. (2026). Improved Localization of CHIME/FRB Far-Sidelobe Fast Radio Bursts. In preparation.

The CHIME/FRB Collaboration + co-authors including Rivera, I. (2026). A Comprehensive Search for Multiwavelength Counterparts to the Second CHIME/FRB Baseband Catalog. In preparation.

Awards

Professor Mercedes T. Richards Award for Excellence in Summer Undergraduate Research — \$2,000

September 2026

SURP, University of Toronto

- Awarded to one SURP student demonstrating outstanding achievement and growth in research, evaluated across technical/research skills, independence of thought, and science communication.
- Selected from 30+ students.

The Roy Alvin Hope Scholarship — \$1,000 <i>Victoria University, University of Toronto</i>	<i>August 2026</i>
• Awarded to students who achieved overall A standing in third year.	
Dunlap Admin's Bestie Award <i>David A. Dunlap Department of Astronomy and Astrophysics</i>	<i>January 2026</i>
• Awarded to consistent, dependable outreach volunteers at the Dunlap Institute. • First undergraduate recipient in the department's history.	
John L. Horwood Memorial Scholarship — \$1,000 <i>David A. Dunlap Department of Astronomy and Astrophysics</i>	<i>August 2025</i>
• Awarded to a student with overall A standing in second year, with preference for astronomy coursework.	
Samuel Beatty Travel Award — \$2,000 <i>University of Toronto Physics Department</i>	<i>January 2025</i>
Admissions Scholarship — \$1,000 <i>University of Toronto</i>	<i>September 2023</i>

Presentations

AST201: Stars and Galaxies — Guest Lecturer <i>University of Toronto</i>	<i>August 2026</i>
Physics Union Undergraduate Research Conference — Talk <i>University of Toronto</i>	<i>March 2026</i>
"One Neat Thing" — Talk <i>University of Toronto</i>	<i>December 2025</i>
Dynamic Radio Sky (DRS) — Flash Talk & Poster <i>McGill University</i>	<i>August 2026</i>
SURP Poster Fair — Poster <i>University of Toronto</i>	<i>August 2026</i>
Canadian Astronomical Society (CASCA) Annual Meeting — Poster <i>Université de Montréal</i>	<i>June 2026</i>
Research Opportunity Program (ROP) Poster Fair — Poster <i>University of Toronto</i>	<i>April 2025</i>

Outreach & Leadership

President, Astronomy & Space Exploration Association (ASX) <i>University of Toronto</i>	<i>April 2025 – Present</i>
• Symposium Director, ASX Symposium (2025–2026) • Events Director, ASX (2024–2025)	
Volunteer, AstroTours & Astro on TAP <i>University of Toronto – Department of Astronomy and Astrophysics</i>	<i>Sept 2024 – Present</i>
External Affairs Director, Women in Science and Engineering (WISE) <i>University of Toronto</i>	<i>April 2024 – April 2025</i>
Volunteer, Girls in STEM Workshop Event <i>University of Toronto – Department of Astronomy and Astrophysics</i>	<i>May 2024</i>
Events Assistant, Work Study Program <i>Dunlap Institute for Astronomy and Astrophysics</i>	<i>April 2024 – August 2024</i>
Host <i>Ontario Science Centre</i>	<i>April 2024 – September 2024</i>

Skills

- **Technical:** Python, NumPy, SciPy, Pandas, Git, LaTeX, command-line workflows. Completed Computational Astrophysics (CTA200, University of Toronto) with a final grade of an A+.
- **Research Areas:** Radio astronomy, time-domain astrophysics, and stellar evolution.
- **Communication & Leadership:** Scientific literature synthesis, research presentations, scientific communication, and leadership experience.
- **Languages:** English; beginner Spanish.